Practical 4

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

# Load dataset from a URL (or use pd.read_csv('boston.csv') if you have the file)
url = "https://archive.ics.uci.edu/ml/machine-learning-databases/housing/housing.data"
names = ['CRIM', 'ZN', 'INDUS', 'CHAS', 'NOX', 'RM', 'AGE', 'DIS', 'RAD', 'TAX',
'PTRATIO', 'B', 'LSTAT', 'MEDV']
df = pd.read_csv(url, delim_whitespace=True, names=names)

# Select one feature (X) and the target (y)
X = df[['RM']]  # Average number of rooms
y = df['MEDV']  # House Price

# Split into Training (80%) and Testing (20%)
x_train, x_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# Create and Train the model
model = LinearRegression()
model.fit(x_train, y_train)

# Make predictions
y_pred = model.predict(x_test)

import matplotlib.pyplot as plt

# Check accuracy (R-squared score)
print("Model Score:", model.score(x_test, y_test))
```

```
# Plot the results
plt.scatter(x_test, y_test, color='blue') # Actual data points
plt.plot(x_test, y_pred, color='red')     # The Regression Line
plt.xlabel("Rooms")
plt.ylabel("Price")
plt.show()
```